

Building the Budget

Project: SkillUp

Team: 3

Date: October 5, 2026

1. Assumptions

Cost-driving assumption	Why it affects cost	Owner
The budget is based on the User Portfolio Management and Analysis Slice	Changes or adding to the slice increases effort and cost	Project Manager
Team size of 5 people	Changes in team causes extra coordination efforts	Project Manager
Apple and Android Apps and Website access to the platform	We need to test and develop platform specific features.	Integration Lead
Platform must support 1k – 2k active users at launch	Lower expected users will result in wasted resources. Having more than expected users will require to scale up.	Technical Lead
6–12-month development timeline	Delays increase labor and costs.	Project Manager

2. Work Packages

Note: 5 People are partially allocated for this slice

Work package	Role	Estimated hours
User Portfolio and Activity Tracking	Developers	70
Progress Tracking and Reporting	Developers	90
Portfolio Customization and Goals	Developers	55
UX Design	UX	40
Testing and QA	QA	50
Total delivery effort		305

3. Estimating Efforts

Method	Optimistic / inputs	Most likely	Pessimistic	Result (hours)
Analogous	144 hrs	180 hrs	216 hrs	180 hrs
Story Points	248 hrs	326 hrs	414 hrs	326 hrs
Bottom Up	Work Package estimates	-	-	305 hrs

Planning value: 305 hours

Explanation: Analogous and Story points are estimation methods from Lesson 9. The middle estimates from lesson 9 were 180 and 326 with an overall range of 144-414hrs. The Bottom-Up estimate is based on planned work and is within the range of Lesson 9 estimates, we are using 305 hours for the planning value.

4. Convert Hours to FTE-Months

Nominal hours per FTE-month: 160 hours Productive hours per FTE-month: 130

$FTE = \text{Hours} / \text{Productive Project Hours}$

Role	Hours	FTE-months
QA	50	0.38
UX	40	0.31
Developers	215	1.65
	Total project hours: 305	Total FTE-months: 2.34

Total FTE-months = 2.34 using rounded role values.

5. Staff Plan

Role	M1	M2	Total
QA	0.10	0.28	0.38
UX	0.25	0.06	0.31
Developers	0.90	0.75	1.65
Total FTE-Months	1.25	1.09	2.34

Staffing table total FTE-months: 2.34

Matches conversion above? Yes

6/7. Calculated Loaded Labor Rate

Loaded rate assumption: \$10,400 per FTE-month, equivalent to \$80 per productive hour based on 130 productive hours per month

Loaded Labor

Loaded rate: \$ 10,400 per FTE-month OR \$80 per productive hour

Role	FTE-months	Loaded rate	Labor cost
QA	0.38	\$10,400	\$3,952
UX	0.31	\$10,400	\$3,224
Developers	1.65	\$10,400	\$17,160
Labor Total	2.34		\$24,336

Non-Labor

Note: Project-wide software licenses and other shared costs are excluded from this slice budget. This section includes only non-labor costs directly attributable to the selected slice.

Item	How computed	Cost
Development and Test hosting	Set up environment for testing and development purposes. (\$150 / Month) * 2 months	\$300
Security & Compliance tools	Data privacy checks, secure API, user data safeguards etc. (\$1,600/month x 2 months)	\$3,200
Training Materials	Onboarding guides, documentation, and resource creation (\$500 x 2 kits)	\$1,000
Non-Labor Total		\$4,500

Type	Total
Labor Total	\$24,336
Non-Labor	\$4,500
Base Build Estimate	\$28,836

8. Cost of Ownership

Annual Operating Cost of the Selected Slice

Operating item	How computed	Annual cost
Production Hosting	Estimated share of production infrastructure for this slice: $\$500/\text{month} \times 12 \text{ months}$	\$6,000
Support & Maintenance	$0.20 \text{ FTE} \times \$10,400/\text{month} \times 12 \text{ months}$	\$24,960
Security and Compliance Services	Estimated slice share: $\$400/\text{month} \times 12 \text{ months}$	\$4,800
Operating Total		\$35,760/year

9. Price the Risks (Contingency Reserve)

$EMV = \text{probability (chance)} \times \text{cost impact}$

Risk	Chance	Cost impact	EMV
Security/Compliance rework	30%	\$4,000	\$1,200
Over-expected user load	25%	\$3,000	\$750
Portfolio and statistics rework	20%	\$2,500	\$500
Contingency reserve			\$2,450

Contingency reserve = sum of EMV = \$2,450

Cost baseline = base build + contingency = \$28,836 + \$2,450 = \$31,286

Management reserve rate: 5% => Management reserve: \$1,564

Total budget request = cost baseline + management reserve = \$32,850

10. Time-Phase

Labor = FTE-Month (Month Specific) * Labor Rate

Month	Labor	Non-Labor	Total	Cumulative
1	\$13,000	\$1,750	\$14,750	\$14,750
2	\$11,336	\$2,750	\$14,086	\$28,836
Total	\$24,336	\$4,500	\$28,836	\$28,836

One-Page Budget Summary

Budget line	Amount
Labor	\$24,336
Non-labor	\$4,500
Base build estimate	\$28,836
Contingency reserve	\$2,450
Cost baseline	\$31,286
Management reserve	\$1,564
Total budget request	\$32,850

Annual operating cost (separate): \$ 35,760 / year

Most important assumption / budget note: The budget is for the Portfolio Management and Analysis slice and assumes a two-month development period. The overall SkillUp project is expected to take 6-12 months, support 1,000-2,000 active users at launch, and the planned staffing and third-party costs remain within the estimated ranges.